

Clone-Based Weighting for Future-Dependent Event Definitions

Hongseok Kim
CSL Behring JSM 2026 · Boston, MA

When diagnosis needs two positives, how do we analyze a single, unconfirmed test?

Introduction

Standard time-to-event analysis requires that, at any fixed time t , one can determine whether the event of interest has occurred by t (Andersen et al. 1993). Some applied event definitions do not meet this requirement—deciding whether the event occurred at time t requires information from later time points. We call these **future-dependent event definitions**.

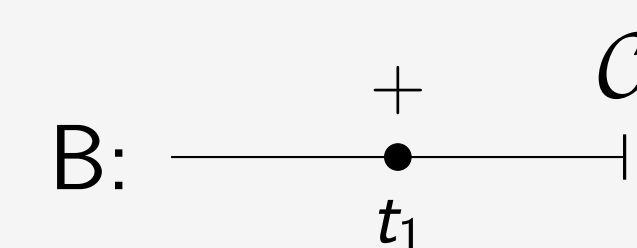
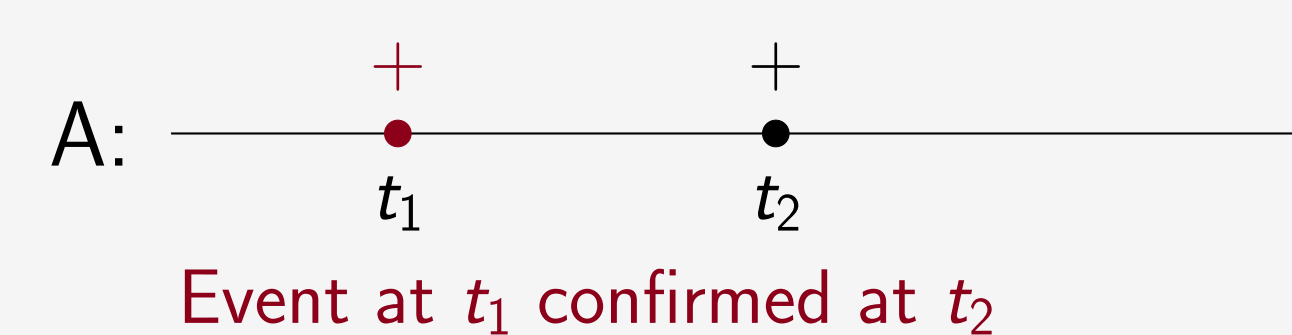
Such definitions typically arise from substantive scientific needs, such as reducing false positives by requiring confirmation, or defining outcomes (e.g., “sustained response”) that inherently span multiple measurements. Without careful handling, they introduce ambiguity in event timing and can result in misleading estimates.

We focus on one common instance: a confirmed diagnosis defined as **two consecutive positive test results**. We propose a clone-based weighting correction for the Kaplan–Meier estimator. The approach extends naturally to other future-dependent event definitions.

Motivation

- Confirmatory diagnoses requiring **two consecutive positive test results** are common in clinical research (Heerspink et al. 2020; Phillips et al. 2016; Saracino et al. 2016).
- Requiring confirmation reduces false positives (Moses et al. 2016; Selvin et al. 2018).
- However, **censored first-positives** (individuals with one positive result and no second test before censoring) create **ambiguity** about whether the event occurred.

$+$: positive test result C : censoring



Whether the event occurred at t_1 is uncertain: it could only be confirmed by a second positive that was never observed.



Scan for code, supplementary materials & references

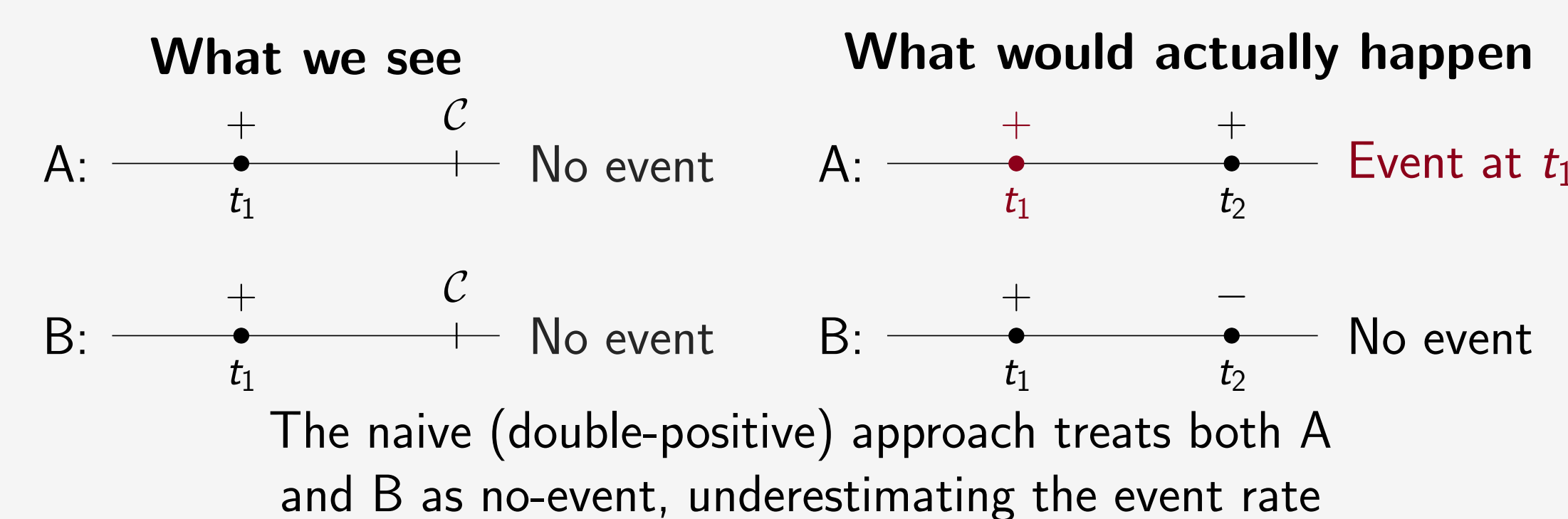
CSL Behring

Methods

Notation

- n_i : risk set size in time interval i
- y_i^* : number of first-positive individuals in interval i who would be confirmed had follow-up continued
- p_i : probability that a first-positive in interval i would be confirmed at the next test

Specification of bias

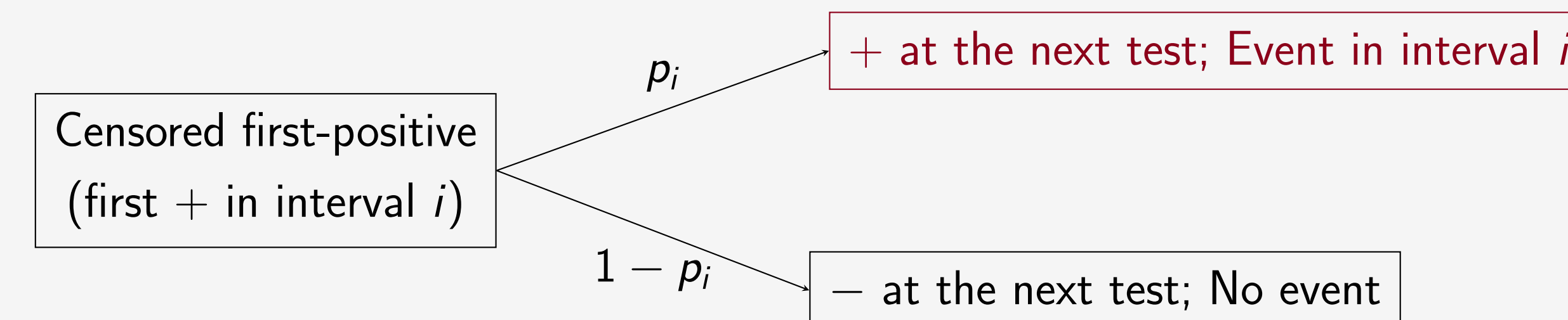


$$\text{bias} \approx \frac{y_i^*}{n_i}$$

The size of this bias is driven by how often a first positive is confirmed at the second test, and by how soon that second test occurs.

Bias correction via cloning and weighting

- Necessity check:** confirm that a future-dependent event definition is truly required.
- Cloning:** For censored first-positives, create two clones representing the two alternative futures.
- Weighting:** Assign each clone a weight equal to the probability of its future: p_i for the event clone, $1 - p_i$ for the no-event clone.
 - p_i may be known, externally estimated, or modeled from data.



► Censored first-positives are split into clones with weights p_i and $1 - p_i$; all other individuals keep weight 1. A standard weighted Kaplan–Meier estimator then recovers the omitted hazard contribution y_i^*/n_i in expectation when p_i is correctly specified.

Simulation setup True disease onset $T_0 \sim \text{Exp}(\lambda = 0.15)$, $n = 1000$ (500 replicates). Each individual has up to $K = 5$ scheduled tests; the gap to the next test is shorter after a positive result. Tests have false-positive rate ε (full settings in the supplementary materials).

Four Kaplan–Meier estimators:

- True:** based on error-free test results (gold standard)
- Single-positive:** event at first observed positive
- Double-positive:** event at first of two consecutive observed positives, censored first-positives left as censored
- Clone-weighted:** as above, with censored first-positives split into weighted clones using the true p_i (proposed)

Results

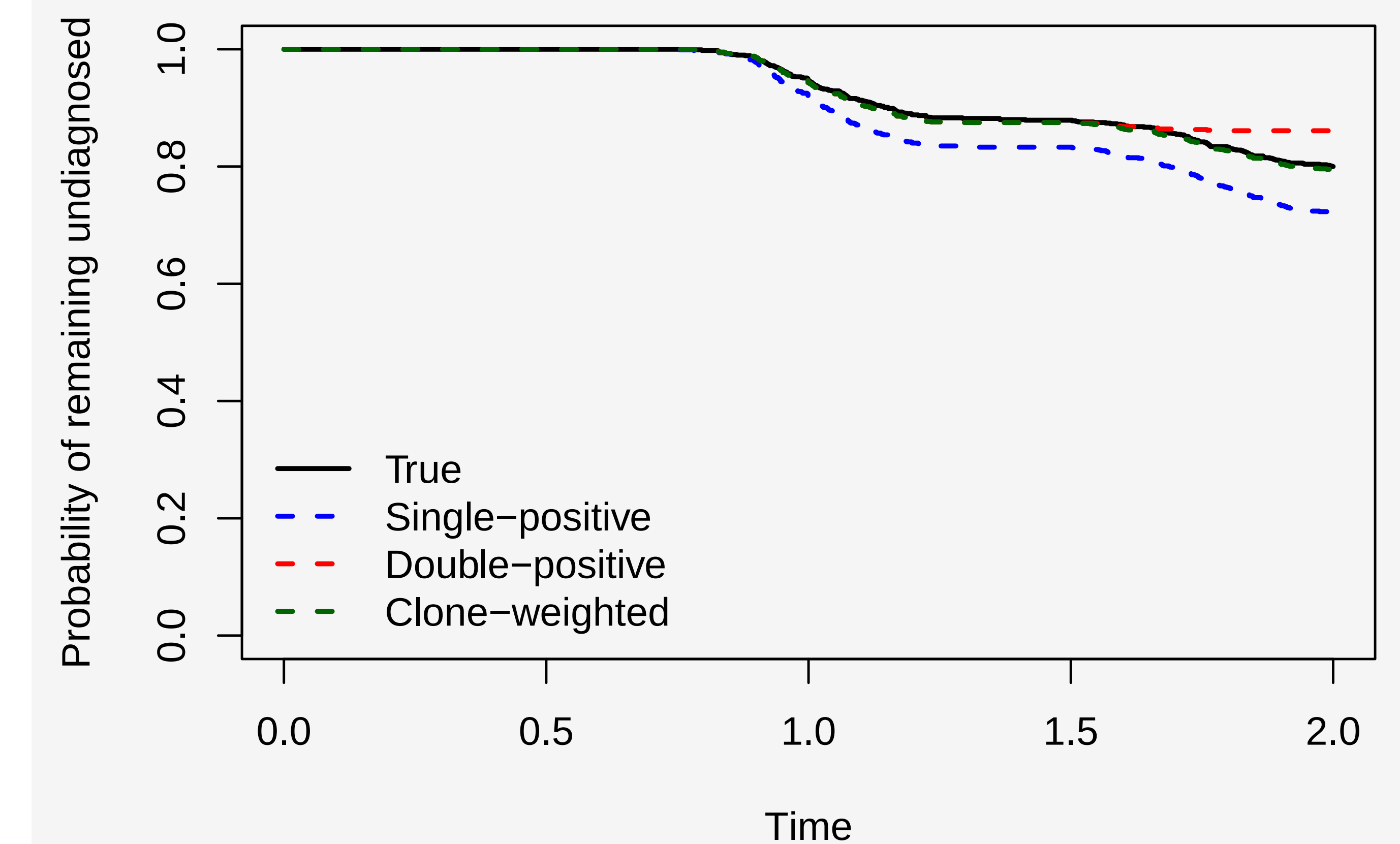


Figure 1: Kaplan–Meier estimates at the baseline setting. The single-positive estimator overestimates the event rate; the double-positive underestimates it; the clone-weighted estimator stays close to the truth.

► Clone weighting accounts for the outcome ambiguity of censored first-positives.

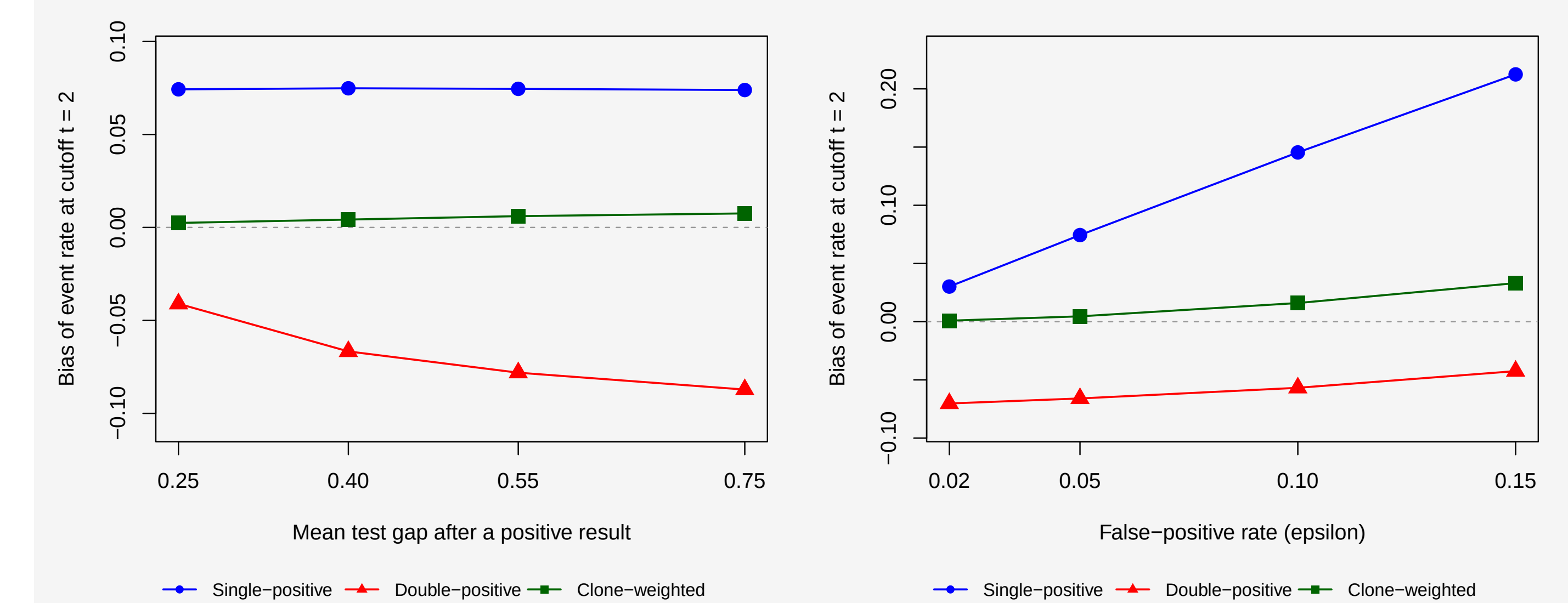


Figure 2: Bias in the event rate at the cutoff. *Left:* widening the gap to the confirming test increases the double-positive estimator’s bias, while the clone-weighted estimator’s bias stays near zero. *Right:* clone weighting reduces bias across false-positive rates, though some bias from false positives remains and grows as the rate rises (correctable when the rate is known; see supplementary materials).

► Clone weighting removes the bias from censored first-positives at any retesting frequency; the bias from false positives remains and grows with the false-positive rate.

Conclusions

► Avoid future-dependent event definitions when you can; when you cannot, clone weighting removes the bias introduced by the naive double-positive approach.